

Propaganda Detection: Technique Classification and Span Identification

1. Introduction

Propaganda detection has become an important Natural Language Processing (NLP) task due to the rapid spread of misleading and emotionally manipulative content across online news platforms and social media. In propaganda, emotional manipulation and persuasion through certain language tricks are key elements used to change people's opinions on the matter, create a political narrative, and persuade people with emotions rather than objective arguments. In order to identify the propaganda content in modern days of a large amount of content, the manual approach has become unfeasible, automatic approaches should be considered.

In this coursework, two different propaganda tasks are studied. The first task considers the classification of propaganda techniques, in which the labelled propaganda span is provided alongside its sentence, and the system is supposed to classify the propaganda technique used among the possible eight types. The second task consists of the identification of propaganda spans in the sentences and then classifying them according to their techniques.

Both problems are inherently difficult since many propaganda methods have common linguistic features, which makes class separation very hard. Propaganda spans detection further needs good accuracy on the token level, where any boundary mistake negatively affects evaluation results. The small size of the dataset also contributes to overfitting risks for transformer-based models.

Several methods were applied and examined for both tasks. In regard to Task 1, a standard TF-IDF with a Logistic Regression model, along with a DistilBERT model, which can learn contextually-based semantic representations, was used. Regarding Task 2, BIO tag labelling methods were developed based on token classification models. The project explored various preprocessing techniques, token-level annotations creation, and assessment via Macro F1-score and sequence labelling measures.

2. Dataset Overview and Preprocessing

2.1 Dataset

The dataset used in this coursework consists of propaganda-labelled sentences in TSV format, where propaganda spans are marked using <BOS> and <EOS> tags. The dataset contains eight propaganda techniques together with a not propaganda class. For Task 1, all not propaganda examples were removed, leaving only propaganda-labelled spans for technique classification. The dataset presents several challenges. Many propaganda techniques share similar linguistic patterns, making class separation difficult, while propaganda span lengths vary significantly, requiring accurate token-level prediction. In addition, the relatively small dataset size increases the risk of overfitting for transformer-based models such as DistilBERT.

2.2 Preprocessing

Several preprocessing steps were applied before training the models. First, the <BOS> and <EOS> markers were used to extract propaganda spans from each sentence. The extracted spans were used for Task 1, while clean sentences without the markers were retained for Task 2 span detection. Missing spans were removed, and dataframe indices were reset for consistency.

For Task 1, only propaganda-labelled examples were retained because the objective was propaganda technique classification. The extracted spans were converted into TF-IDF features and transformer token representations for the TF-IDF and DistilBERT models. For Task 2, full sentence context was preserved for token-level span detection. BIO labels were generated using B (beginning), I (inside), and O (outside) tags. Padding and truncation were applied during tokenisation to create fixed-length input sequences for DistilBERT training.

3. Method

3.1 Task 1A: TF-IDF + Logistic Regression

In the first approach, TF-IDF feature extraction was combined with a Logistic Regression classifier as a traditional machine learning baseline. The TF-IDF converted propaganda spans into numerical feature vectors based on word importance within the dataset. Unigram and bigram features were used with `ngram_range = (1,2)` to capture both individual words and short propaganda-related phrases. The vectoriser also used a maximum vocabulary size of 10,000 features together with English stop-word removal.

The Logistic Regression classifier was trained using balanced class weights to reduce the impact of class imbalance and `max_iter = 1000` to ensure convergence during training. The best-performing configuration used span-only representations, achieving a Macro-F1 score of 0.3814. The span-only representation outperformed sentence-context representations because propaganda techniques could often be identified directly from the extracted spans, while additional sentence context introduced noisy and less relevant information.

3.2 Task 1B: DistilBERT Classification

The second method made use of a transformer-based DistilBERT model to classify propaganda techniques. While the TF-IDF model ignores context in favour of frequency, the DistilBERT model creates contextual embeddings for words wherein word meaning is dependent on other words surrounding it. This helps the model to learn semantic and contextual interactions between words, something that could help immensely in propaganda, wherein many propaganda techniques are based upon subtle emotional and contextual cues.

The extracted propaganda spans were tokenised using the DistilBERT tokenizer with padding and truncation applied to create fixed-length input sequences suitable for batch training. The tokenised inputs were then passed through the DistilBERT encoder, which produces contextual semantic representations for each token. A linear classification head was added on top of the transformer output to predict one of the eight propaganda technique classes.

As mentioned before, the DistilBERT model was chosen for its efficiency and contextual awareness as opposed to other transformer models like BERT, RoBERTa, etc., that have a large number of layers. Performance for DistilBERT was evaluated via the Macro F1-score for a better evaluation of the propaganda. The DistilBERT model obtained a Macro-F1 score of **0.5207**, higher than that obtained from the TF-IDF model.

3.3 Task 2A: Pipeline BIO Tagging

In the first strategy employed for Task 2, we made use of the pipeline architecture where sentence-level propaganda detection was followed by BIO tagging to detect span boundaries. First, a binary classifier was built that classified each sentence into propagandist and non-propagandist categories. Binary labels were formed in a manner such that sentences containing propaganda were marked with label 1 while the other non-propagandist sentences received label 0. Sentence-level propagation classification was done using a sentence vectorizer coupled with a logistic regression classifier. We employed a TF-IDF vectorizer with unigrams and bigrams representation without English stop words. Sentence text data underwent preprocessing before being used in model training and evaluation.

Propaganda-negative sentences were ignored in the next step of the pipeline process. For propagandist sentences, the BIO tagging method was utilised to assign each sentence with B (beginning of a propaganda span), I (inside a propaganda span) or O (outside propaganda span) labels. The span identification task thus became a sequence labelling problem. It is important to perform sequence labelling since different propaganda spans contain different numbers of tokens. Thus, it was critical to apply label prediction at the token level.

3.4 Task 2B: DistilBERT Token Classification

For the second task, a two-approach DistilBERT-based token classification model was applied for detecting propaganda spans as well as classifying techniques. BIO tags were created for each sentence according to the following convention: B-technique, I-technique, O. This means that the task could be considered as a token-level sequence labelling one. The sentences were tokenised with DistilBERT tokenizer, padded and truncated so that fixed-size input sequences were created. Given the possibility of splitting words into subwords in transformer tokenisation, label alignment needed to be done in order to make sure that BIO tags would align with the created tokens. In case a word was split into several subwords, continuation tokens would inherit the I-tag.

The model was built on the basis of the `AutoModelForTokenClassification` class. In this case, DistilBERT was used for generating token embeddings while a token classification head produced BIO tags for each token. AdamW optimizer with gradient clipping was used for training the model. This made it possible to use the model for learning context-aware token representations and performing propaganda span detection and technique classification at the same time. The BIO tagging is especially important since propaganda spans can have various lengths.

3.5 Training Strategies

Several training methods have been employed to enhance model robustness and generalisability during experiments. For the TF-IDF + Logistic Regression classifier, class balancing was implemented to account for imbalanced classes among different types of propaganda tactics. The maximum iteration parameter was extended to 1000 to ensure convergence during the optimisation process. For the DistilBERT models, the padding and truncation strategy was applied to create fixed-length input sequences during tokenisation to accommodate batch training. The AdamW algorithm was chosen for optimisation, with a learning rate of $1e-5$, alongside a learning-rate scheduler, which is linear in nature. Gradient clipping, with a norm value of 1.0, was implemented to stabilise gradient updates.

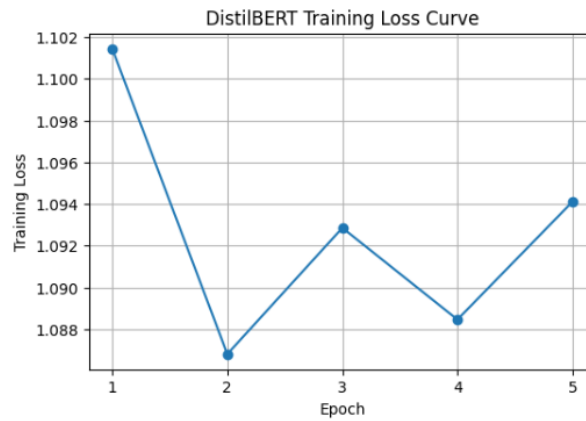


Figure 1: Training loss curve

For Task 2, label alignment was conducted carefully based on the BIO scheme to accurately align word-level labels with transformer subwords. In addition, validation sets were utilised during training to assess model performance based on Macro F1-score and sequence-level F1 scores.

4. Hyper-parameter Settings

Several hyperparameters were selected to improve model performance and training stability.

For the TF-IDF + Logistic Regression model, `max_features = 10000` was used to balance vocabulary coverage and computational efficiency. The `ngram_range = (1,2)` allowed the model to capture both individual words and short propaganda-related phrases. English stop-word removal reduced less informative words, while `max_iter = 1000` ensured the Logistic Regression model converged during training. Balanced class weights were applied to reduce the effect of class imbalance across propaganda techniques.

For the DistilBERT classification model, `distilbert-base-uncased` was selected because it provides strong contextual understanding while remaining computationally efficient. A maximum sequence length of 128 was sufficient because most propaganda spans were short. A batch size of 8 balanced GPU memory usage and training efficiency, while a learning rate of $1e-5$ was used to ensure stable transformer fine-tuning. Training was performed for 5 epochs to reduce underfitting without excessive overfitting. For Task 2 token classification, a larger

sequence length of 256 was used because full sentences and BIO labels required more contextual information. A smaller batch size of 4 was used due to higher memory requirements for token-level classification. Gradient clipping with $\text{max_norm} = 1.0$ was applied to prevent unstable gradient updates during transformer training.

5. Results and Evaluation

Model performance was evaluated using the macro F1-score for Task 1 and span-level F1-score for Task 2. Macro F1-score was selected for propaganda technique classification because it evaluates all classes equally and reduces the impact of class imbalance. For Task 2, span-level F1-score from the sequeval library was used because propaganda span detection requires accurate token-level BIO predictions and exact span boundary matching.

5.1 Task 1: Results

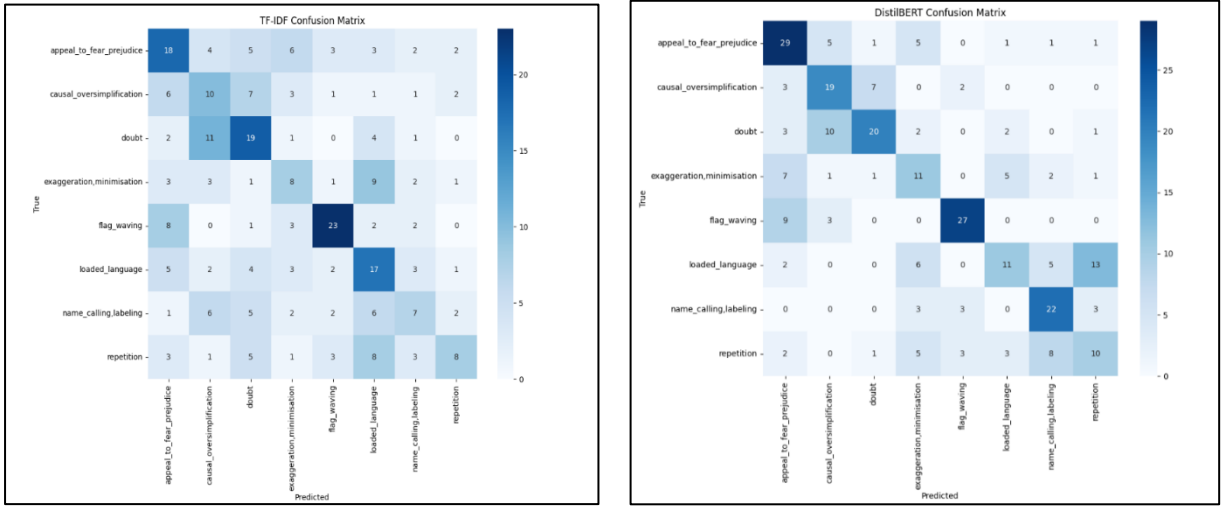


Figure 2: Confusion matrix for the TF-IDF + Logistic Regression classifier & DistilBERT

The TF-IDF confusion matrix shows that the model frequently confused semantically similar propaganda techniques such as `causal_oversimplification`, `doubt`, and `loaded_language`. Although techniques such as `flag_waving` achieved relatively strong performance due to distinctive lexical patterns, the overall confusion between classes indicates the limitations of sparse lexical representations for fine-grained propaganda classification. The DistilBERT confusion matrix demonstrates improved class separation compared with the TF-IDF baseline, with stronger diagonal prediction patterns across most propaganda techniques. The contextual semantic representations learned by DistilBERT enabled the model to better distinguish overlapping propaganda categories, resulting in higher overall Macro F1 performance.

Table 1 compares the performance of the TF-IDF + Logistic Regression model and the DistilBERT classifier.

The DistilBERT model achieved substantially better performance than the TF-IDF baseline, demonstrating the importance of contextual semantic representations for propaganda detection. While the TF-IDF model relied mainly on lexical frequency patterns, DistilBERT was better able to capture contextual and semantic relationships between propaganda spans.

Model	Macro-f1
TF-IDF + Logistic Regression	0.3814
DistilBERT	0.5207

Table . Marco f1 scores for task 1 models

5.2 Task 2: Results

Two BIO tagging approaches were evaluated for propaganda span detection and technique classification.

Model	Span-Level F1
Generic BIO Tagging	0.68
Technique-Aware BIO Tagging	0.029

Table 2. BIO tagging

The generic BIO tagging pipeline achieved substantially higher span-level performance than the technique-aware BIO tagging model. The direct token classification approach struggled because the number of BIO labels increased significantly when propaganda techniques were incorporated directly into the labels, making sequence prediction more difficult with the limited dataset size.

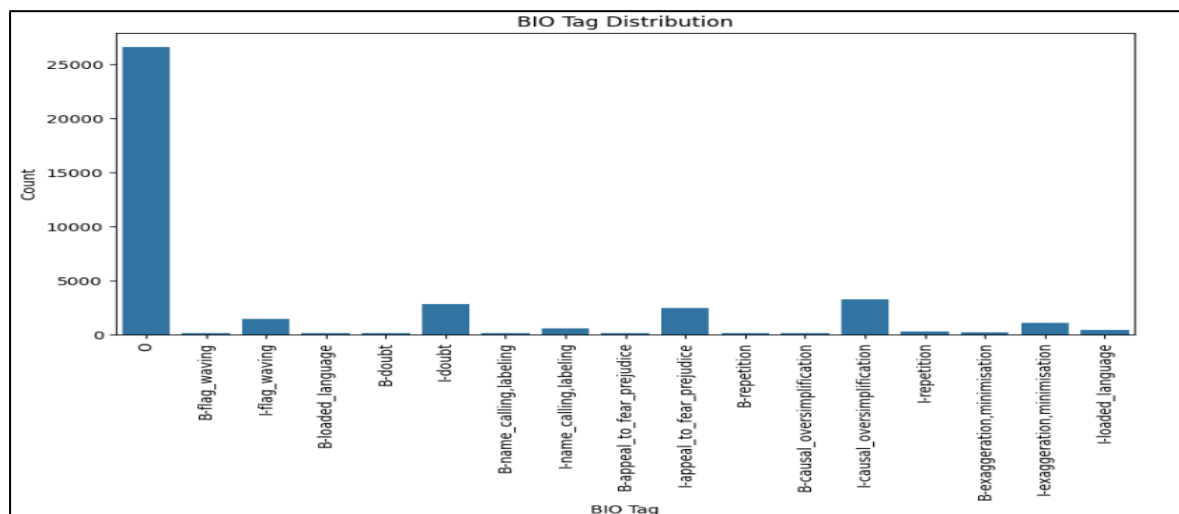


Figure 3: BIO tag distribution graph

The BIO tag distribution shows a strong imbalance between the O label and propaganda-related BIO tags. Most tokens in the dataset belong to the O category, while comparatively fewer tokens are labelled using B- and I- propaganda tags. This imbalance makes token-level propaganda detection difficult because the model is exposed predominantly to non-propaganda tokens during training. In particular, longer techniques such as causal_oversimplification and appeal_to_fear_prejudice contain many I-labels, whereas shorter techniques such as loaded_language and repetition contain relatively few labelled tokens. The imbalance likely

contributed to the poor performance of the technique-aware BIO tagging model, which frequently predicted O labels instead of propaganda spans.

6. Analysis

The experimental results revealed several common error patterns across both propaganda classification and span detection tasks. For Task 1, the TF-IDF model frequently confused semantically similar propaganda techniques such as `causal_oversimplification`, `doubt`, and `loaded_language`. Since TF-IDF relies primarily on lexical frequency information, the model struggled when different propaganda techniques shared similar vocabulary or emotional expressions. Although `flag_waving` achieved relatively strong performance due to its distinctive lexical patterns, more context-dependent techniques were harder to classify correctly. The DistilBERT classifier improved overall performance by using contextual semantic representations, but errors still occurred for techniques with overlapping rhetorical meaning. In particular, `loaded_language` and `repetition` remained difficult because short propaganda spans often lacked sufficient contextual information for reliable classification.

	Token	True_Label	Predicted_Label	Column2
0	Mostly	O	O	✓
1	because	O	O	✓
3	country	I-causal_oversimplification	I-flag_waving	✗
4	would	I-causal_oversimplification	O	✗
5	not	I-causal_oversimplification	O	✗
6	last	I-causal_oversimplification	O	✗
7	long	I-causal_oversimplification	O	✗
8	without	I-causal_oversimplification	O	✗
9	an	I-causal_oversimplification	O	✗
10	outside	I-causal_oversimplification	O	✗
11	high	I-causal_oversimplification	O	✗
12	IQ	I-causal_oversimplification	O	✗
13	elite	I-causal_oversimplification	O	✗
14	to	I-causal_oversimplification	O	✗
15	run	I-causal_oversimplification	O	✗
16	the	I-causal_oversimplification	O	✗
17	country	I-causal_oversimplification	I-flag_waving	✗
18	.	O	O	✓

Table 3: Token Prediction Error Table

For Task 2, the technique-aware BIO tagging model performed poorly compared with the generic BIO tagging pipeline. The BIO tag distribution showed that most tokens in the dataset were labelled as O, creating severe class imbalance during training. As a result, the token classification model frequently predicted non-propaganda labels even when propaganda spans were present. The token-level prediction examples further showed that the model struggled particularly with long propaganda spans such as `causal_oversimplification`. In many cases, the model predicted O labels for entire propaganda fragments, indicating difficulty learning exact span boundaries and technique-specific BIO transitions simultaneously. The limited dataset

size and large number of BIO labels also contributed to weak generalisation performance for token classification.

7. Conclusion

This coursework investigated propaganda technique classification and propaganda span detection using both traditional machine learning and transformer-based approaches. For Task 1, the DistilBERT classifier achieved better performance than the TF-IDF + Logistic Regression baseline because contextual transformer embeddings captured semantic relationships more effectively. For Task 2, BIO tagging approaches were used for token-level propaganda span detection. The experiments showed that propaganda span identification is considerably more challenging due to BIO label imbalance, limited training data, and exact span boundary prediction difficulties. Overall, the results demonstrated that transformer-based contextual models are more effective than traditional lexical feature-based methods for propaganda detection tasks.

8. Further work

Several improvements could further enhance propaganda detection performance. Larger transformer architectures such as RoBERTa or DeBERTa could be explored to improve contextual understanding and classification accuracy. Additional data augmentation techniques and larger annotated datasets may help reduce overfitting and improve generalisation, particularly for token-level span detection. Future work could also investigate CRF-based decoding layers to improve BIO sequence consistency and span boundary prediction. Furthermore, ensemble approaches combining classical machine learning and transformer models could be extended to token classification tasks. Explainability methods could also help better understand transformer-based propaganda predictions.

References

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AI-based tools were used to assist with code development and debugging purposes.